

* Simon Kuznets proved using data from 1869 - 1929
 that APC does NOT ↓ with income. { U.S.A & U.K data }
 (known as Kuznet's Puzzle.

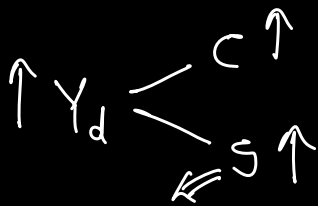
Oct. 29 1929 $\Rightarrow$ 1932 $\rightarrow$ unemployment $\sim$ 25%
 Real GDP ↓ by 30%
 $\rightarrow$ American stock fell to one-tenth of its value.

RECAP:

$$C = f(Y_d) ; f' > 0$$

current real consumption is an increasing function of current real disposable income.

If tax ↓ ; $\uparrow Y_d = Y + TR - T_x \downarrow$
 $\Downarrow$
 disposable income goes up. $\Rightarrow$ of the household sector)
 $\Rightarrow$ consumption goes up (as $C = f(Y_d) ; f' > 0$)



Note: C is the largest component of aggregate demand.
 "Output is demand determined."

$\Rightarrow$ C is the largest component of GDP.

$S = I$
 Saving transforms into Investment

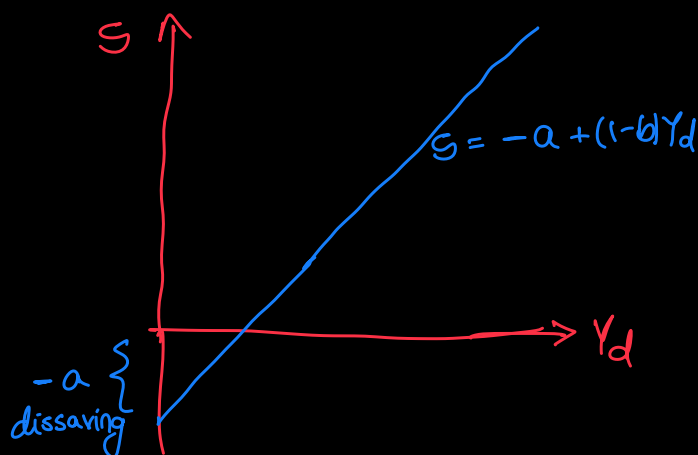
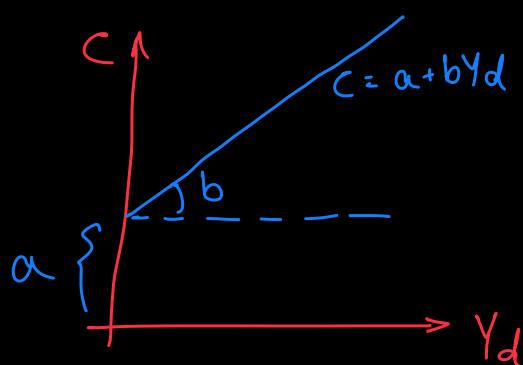
$G + TR - T_x \Rightarrow$ government budget deficit $\Rightarrow G + TR > T_x$

* subsidy $\Rightarrow$ negative tax

$\rightarrow$ If tax rate $\uparrow \Rightarrow Y_d \downarrow \Rightarrow C \downarrow \Rightarrow$ aggregate demand $\downarrow$
 $\Downarrow$
 Output $\downarrow$.

Keynes $\rightarrow$ classicists are fundamentally flawed.

$\Downarrow$ consumption is the largest component of aggregate demand.



(1942)
* Kuznet's said APC remained remarkably stable
as income $\uparrow$ (1869 - 1929) $\Rightarrow$ DATA of U.S, and U.K

(1955)
* Goldsmith verified Kuznet's theory.

Secular Stagnation Hypothesis:

Franklin D Roosevelt $\rightarrow$ introduced new bill.
 $\Rightarrow$ they launched WPA

America did not go into stagnation post WWII

Huge Unemployment
 $\Rightarrow$ low income level
 $\Rightarrow$ low demand for goods & services
 $\Rightarrow$ low output.

If Keynes was true, then $\leftarrow$
American Economy should have gone into stagnation
after WWII as APC $\downarrow \Rightarrow \frac{C}{Y_d} \uparrow \downarrow$

Franco Modigliani

Robert Ando
Richard Brumberg

⇒ Students.

} trying to solve
Kuznet's Puzzle.

(1936)
* Keynes: Income is the primary determinant of consumption

$$C = f(Y_d, \dots)$$

↓
secondary components.

↓ in the short run, this is
correct, because
correlation coefficient
≈ 0.99.

{ Keynes was thinking in the short-run. }

* In the long-run, we all are dead → Keynes.

Modigliani: The secondary determinant is WEALTH.
(↓ Assets)

Objective is smoothest possible consumption pattern.

LCH ⇒ Life Cycle Hypothesis.

$$C = f(\text{Income}, \text{Wealth})$$

objective: smooth consumption over lifetime.

$$\frac{W + RY}{T}$$

W → wealth accumulated
R → working years
Y → income from job

$$= \left(\right) = \frac{W + RY}{T}$$

T → life expectancy

for all practical purposes, $R < T$

$$C = W \left(\frac{1}{T} \right) + \left(\frac{R}{T} \right) Y$$

Note: $R < T$

Both wealth & income have impact on consumption

$$\Rightarrow \boxed{C = \alpha W + \beta Y}$$

$$\alpha < \beta$$

$$\frac{\partial C}{\partial W} = \alpha ; \quad \frac{\partial C}{\partial Y} = \beta \Rightarrow MPC_{Income} > MPC_{Wealth}$$

$$\alpha = \frac{1}{T} > 0 ; \quad \beta = \frac{R}{T} > 0 \text{ and } \beta < 1$$

$$APC = \frac{C}{Y} = \alpha \left(\frac{W}{Y} \right) + \beta \Rightarrow MPC_{Income}$$

Wealth cannot be accumulated in SHORT RUN.

As $Y \uparrow$ in the long-run, wealth also increases. (accumulated)
 $\Rightarrow \frac{W \uparrow}{Y \uparrow} \sim$ remains constant $\Rightarrow$ reflected in Kuznet's finding in long-run.

Milton Friedman : $Y(\text{Income})$
 Permanent Income $\Rightarrow$ SALARY $\rightarrow Y^P$
 Transitory Income $\Rightarrow$ winning KBC. $\rightarrow Y^T$

$$C = \alpha Y^P ; \alpha > 0$$

$$\Rightarrow APC = \frac{\alpha Y^P}{Y} = \frac{\alpha Y^P}{Y^P + Y^T}$$

increase (or)
decrease

$\Rightarrow$ APC can move in 2 different directions

FTA : Free Trade Agreement

As long as it is CLOSED economy, the multipliers depend only on MPC or MPS.

But once we open the economy, there it depends on M' and S' .

→ no international trade

* CLOSED ECONOMY multiplier $>$ OPEN ECONOMY multiplier

$$\left(\frac{1}{1-MPC} \text{ or } \frac{1}{MPS} \right) > \left(\frac{1}{(1-MPC)+MPM} \right)$$

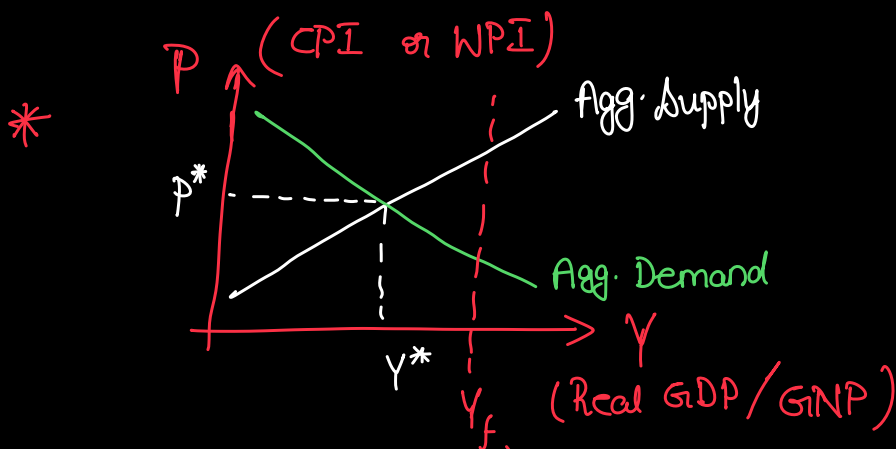
$\{ MPM \sim \text{marginal propensity to import} > 0 \}$

* CLOSED ECONOMY :- single leakage : savings

OPEN ECONOMY :- two leakages : savings & imports.

Why OPEN ECONOMY ?

Strategic Interdependence



$$Y^* < Y_f$$

Underemployment Equilibrium.

$$Y = C + I + G + (X - M)$$

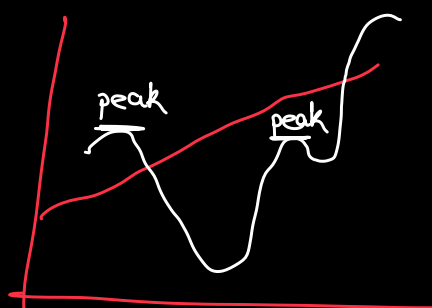
part of fiscal policy

aggregate demand

$$Y_d = C + I + (G + T_R - T_x) + (X - M)$$

fiscal poli

*

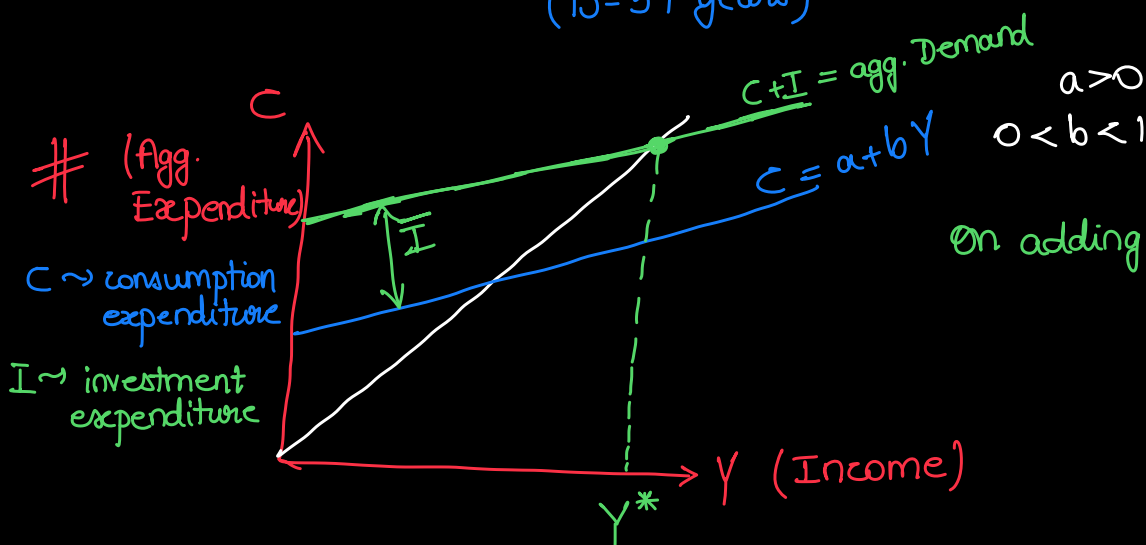


potential output / full employment output
 $\Rightarrow$ MAX sustainable output.

$$\# \text{ LABOUR FORCE} = \text{No. of Employee} + \text{No. of unemployed}$$

$$\text{Unemployment Rate} = \frac{\text{No. of unemployed}}{\text{LABOUR FORCE}} \times 100$$

$$\text{Labour-Force Participation Rate} = \frac{\text{LABOUR FORCE}}{\text{Adult Population (15-59 years)}} \times 100$$



Keynesian Cross Diagram

$$S - I = (G + TR - T_x) + (X - M)$$

Saving Investment Gap.

Budget deficit / surplus

Trade surplus / deficit